MEDISIM

Multimodal Diagnostics & Agentic Triage with Human-In-The-Loop Evaluation

FINAL PROJECT REPORT

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AGENDA

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Problem & Motivation

Clinical dilemma of AI hallucinations and project goals

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Technical Architecture

Vision-Language Fusion and Dual-Axis pipeline

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Quantitative metrics, errors and qualitative findings

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Conclusion & Future

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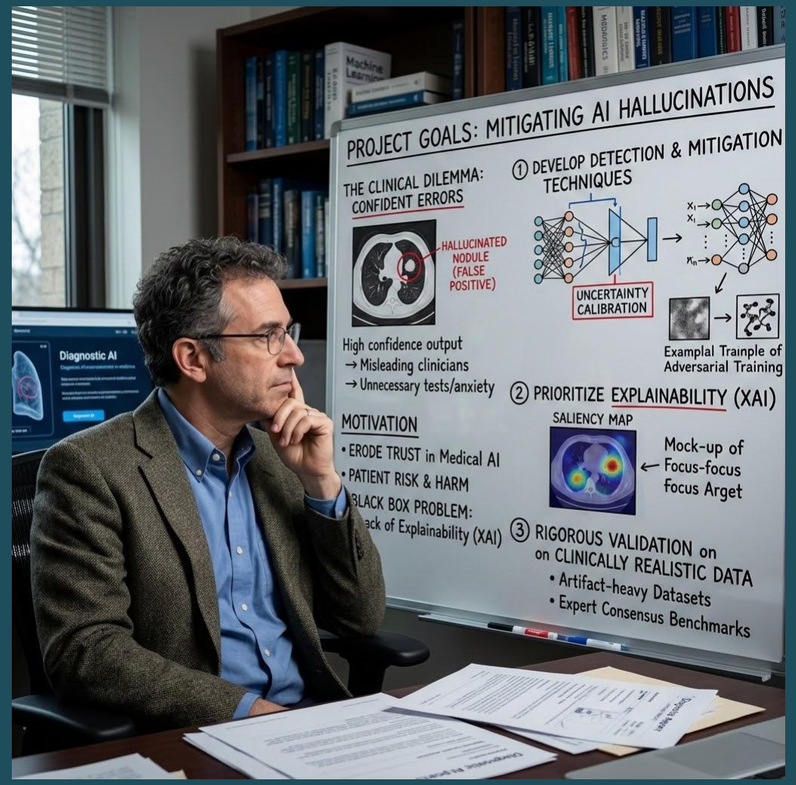
01

Problem & Motivation

The clinical dilemma of AI hallucinations in medical diagnostics and our project goals



Problem Statement & Project Goals

The Clinical Dilemma

Hallucination Risk

Single-agent LLMs hallucinate medical advice without deterministic grounding

Decoupled Pipelines

Vision and text pipelines operate independently, leading to unsafe clinical reasoning

Trust Deficit

No verification mechanism to catch AI errors before reaching patients

Safety Gap

Lack of human-centered evaluation for clinical deployment

Project Goals

Multimodal Fusion & Multi-Agent Triage

Build deterministic Vision + Language fusion pipeline for diagnostic confidence

Implement agentic triage: Nurse, Doctor, Fact-Checker roles

HITL Evaluation

Human-In-The-Loop evaluation for clinical safety and trust

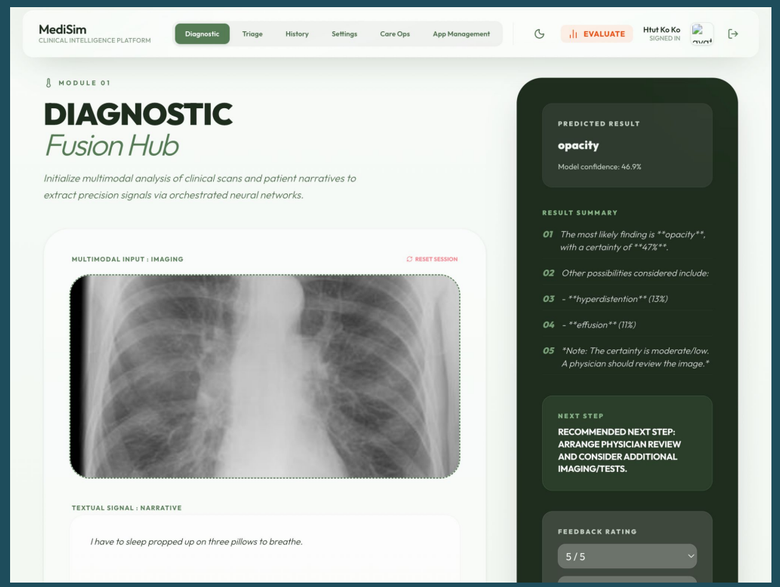
Deployment

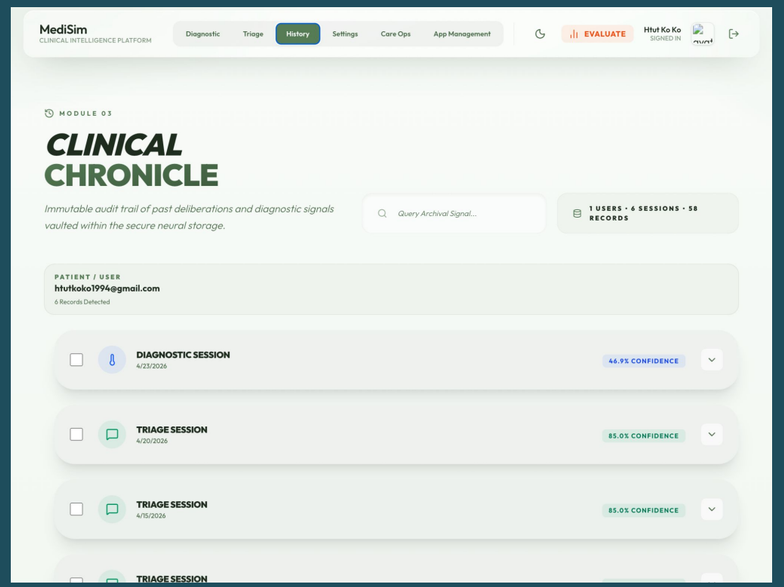
Deployable React/FastAPI ecosystem with real-time telemetry

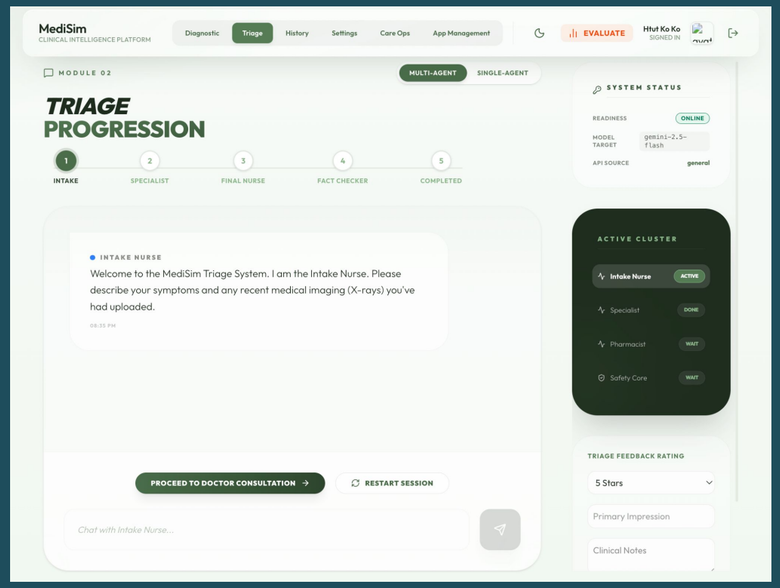
02

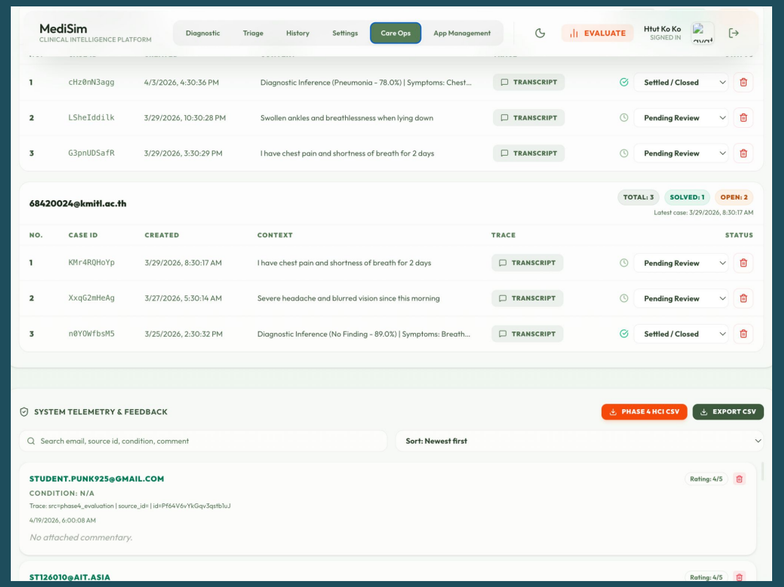
Technical Architecture

Vision-Language Fusion pipeline and Dual-Axis system design









Phase 3: Vision-Language Fusion

Deterministic Backbone

Vision Encoder: ResNet-18 → Chest X-Rays → 128-dim latent

Text Encoder: Bio ClinicalBERT → symptoms → 128-dim

Fusion: Concatenated latent → classification head

Key Result

+5.04% accuracy leap

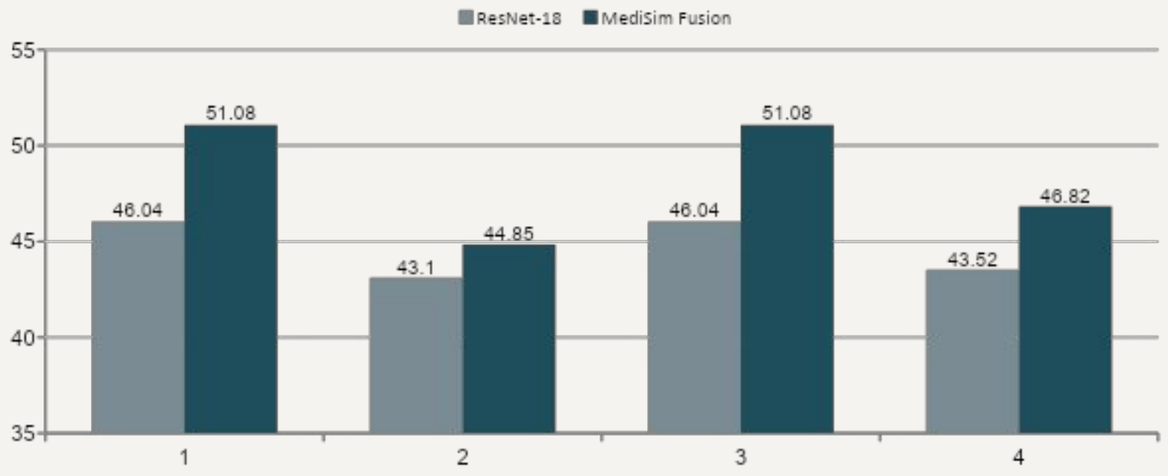
MediSim Fusion vs. ResNet-18 Unimodal on IU X-Ray dataset

Fusion combines visual features from chest X-rays with semantic symptom representations for more robust pathology classification

Performance Comparison

Model Pipeline	Acc	Prec	Rec	F1
ResNet-18	46.04%	43.10%	46.04%	43.52%
MediSim Fusion	51.08%	44.85%	51.08%	46.82%

Metrics Visualization



System Architecture: Dual-Axis Pipeline

AXIS A — Deterministic Analysis

Fused Image + Text Pipeline
ResNet-18 + Bio ClinicalBERT → 128-dim fusion

Explicit Diagnostic Confidence
Probability scores across 15 pathologies

Firebase EMR Storage
Patient records for cross-referencing

↓

Deterministic probabilities ground all generative outputs

AXIS B — Generative Agentic Triage

Triage Nurse
Empathetic data intake — collects symptoms, medical history, and patient concerns

Specialist Doctor
Clinical reasoning — synthesizes fused data for diagnostic recommendations

Fact-Checker
RAG safety gate — cross-references EMR before final output, intercepts unsafe advice

03

Human-in-the-Loop Evaluation

Experimental methodology and HCI evaluation framework

MediSim: Phase 4 Human-in-the-Loop Evaluation Form

Participant ID: P19 — Email: drkhinsu1994@gmail.com

Part 1: Participant Demographics

☐ **Professional Background:** ☒ Clinical/Medical Practitioner ☐ Computer Scientist/Developer ☐ Non-Expert User

☐ **Familiarity with AI Tools (1-5):** 4

Part 2: Multimodal Pipeline Usability (ResNet-Transformer)

Question	1	2	3	4	5
1. Multimodal diagnostic confidence % helped me trust the system.	○	○	○	○	•
2. Uploading combined data (Image + Text) felt intuitive.	○	○	○	○	•
3. Categorical outputs accurately mapped the subjective symptoms.	○	○	○	○	•

Part 3: Multi-Agent Triage Interaction (Heuristic Evaluation)

Question	1	2	3	4	5
4. **Empathy**	○	○	○	○	•
5. **Role Conformity**	○	○	○	○	•
6. **Latency Tolerance**	○	○	•	○	○
7. **Clinical Safety**	○	○	○	○	•

Part 4: Open-Ended Qualitative Feedback

1. Biggest danger of Single-Agent?
"From a clinical standpoint, the single agent cannot differentiate contraindications. It prescribed treatments conflicting with a patient's documented allergy history, which would be catastrophic in a real setting."

2. Fact-Checker module trust?
"As a clinician, seeing the system cross-reference the patient's EMR before generating advice is exactly the safeguard that makes AI trustworthy in a healthcare context. Excellent design."

HITL Methodology & Evaluation Framework

Experimental Design

N = 26 Participants
Clinical Practitioners, Developers, and Non-Experts

Deployment Platform
Fully deployed React/FastAPI ecosystem on Hugging Face

Telemetry
Aggregated interaction vectors (1-5 Likert scales) to Weights & Biases (WandB)

Replaced raw computational metrics with real human trust markers

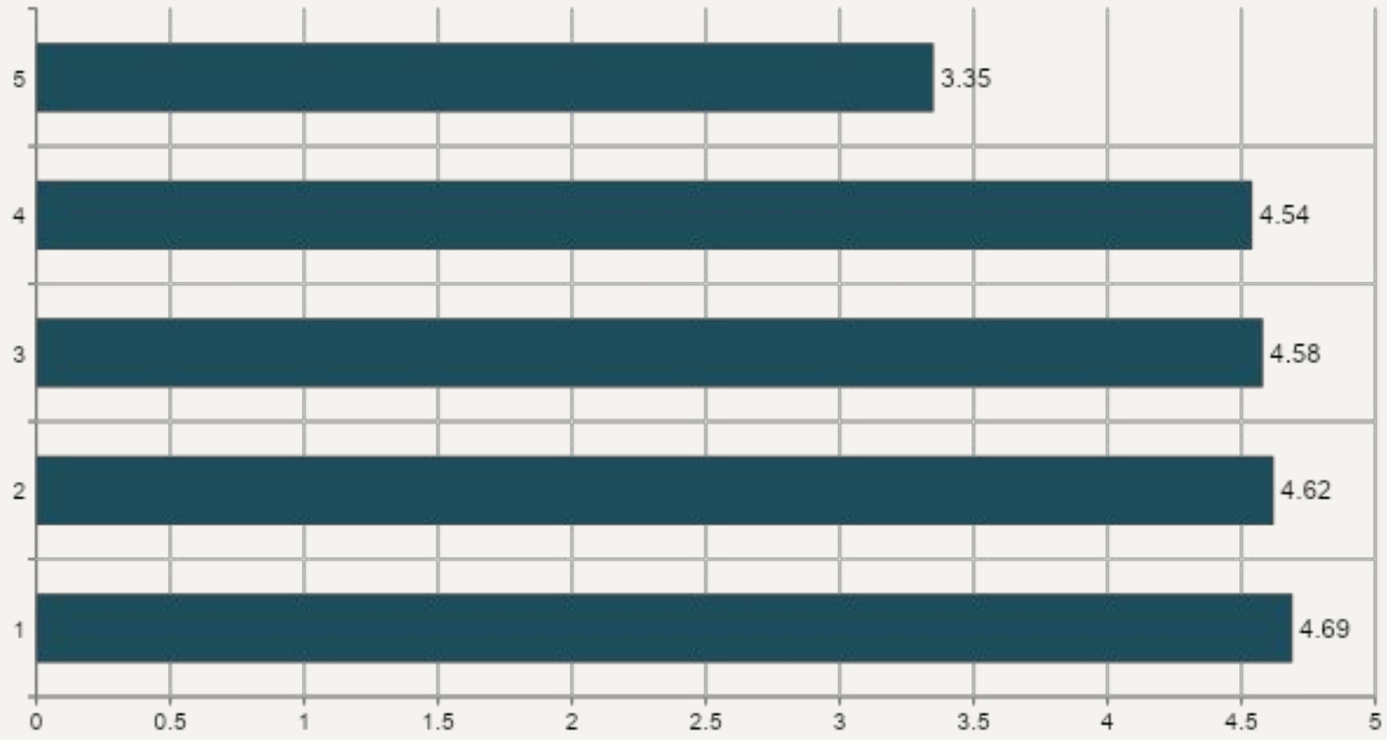
Survey Instrument (4 Parts)

1. **Demographics**
   Role, AI familiarity, clinical background
2. **Multimodal Pipeline Usability**
   UI intuitiveness, trust in % metrics
3. **Multi-Agent Triage Interaction**
   Nurse empathy, specialist conformity, latency, fact-checker safety
4. **Qualitative Feedback**
   Open-ended vulnerability assessment

Quantitative Results: HCI Usability & Trust

Global System Safety & Trust Rating **92.7%**

Heuristic Dimension Ratings (N = 26)



Detailed Metrics

Dimension	Mean	Std Dev
Multimodal Trust	4.58	0.50
Nurse Empathy	4.54	0.51
Specialist Conformity	4.62	0.50
Fact-Checker Safety	4.69	0.47
Latency Tolerance	3.35	0.63

Key Insight: Fact-Checker safety scored highest, validating the RAG verification step. Latency remains the primary bottleneck.

Error Analysis & Limitations

SYSTEM LIMITATIONS

Image Ambiguity

Infiltration frequently confused with Pleural Effusion due to visual overlap in frontal X-rays

Mild Atelectasis misclassified as "Normal"

Negation Handling

Symptoms like "no chest pain" triggered false positive probabilities for pain-related pathologies

Root cause: bag-of-words attention flaws in ClinicalBERT

HUMAN INTERACTION ERRORS

Automation Bias

Non-experts overly trusted initial low-confidence outputs

Example: trusted "Normal" at only 48% confidence

Latency Fatigue

Users abandoned chat prematurely (-8.5s delay) before Fact-Checker could verify

Symptom Ambiguity

Highly emotional inputs from non-experts degraded Triage Nurse extraction accuracy

Qualitative Insights & User Feedback

Hallucination Mitigation

Participants identified all intercepted unsafe advice instances

100% interception rate

Architectural Transparency

Trust generated by "visible verification" steps in the UI

UX design as critical as model accuracy

Latency Bottleneck

Multi-agent chaining is computationally expensive

Primary area: token streaming optimization

USER TESTIMONIALS

"

Seeing the system cross-reference the EMR before generating advice is exactly the safeguard that makes AI trustworthy... excellent design.

P19 — Clinical Practitioner · Safety 5/5

"

The first AI (Nurse) felt warm. Seeing a clear red warning message instead of the AI just giving wrong advice made me feel protected.

P24 — Non-Expert User · Empathy 5/5

Conclusion & Future Work

ACCOMPLISHMENTS

Bridged Vision-Language Models

Successfully integrated ResNet-18 + Bio ClinicalBERT for multimodal diagnostics

Eliminated High-Risk Hallucinations

Deterministic RAG safety gates with 100% interception rate

Full Deployment Ecosystem

React/FastAPI on Hugging Face with WandB telemetry

92.7% Global Safety & Trust Rating

Deterministic probabilities strictly ground generative outputs — the core philosophy validated

FUTURE POSSIBLE WORK

ViT-BERT Fusion

Transition to Vision Transformer for localized quadrant attention on X-rays

ASR Integration

Integrate Whisper-v3 for real-time vocal triage and hands-free interaction

Latency Optimization

Asynchronous token streaming to reduce multi-agent chain delays

Broader Deployment

Scale beyond chest X-rays to other imaging modalities

THANK YOU

https://shadowsilence-medisim.hf.space/

github.com/shadowsilence94/MediSim